

Erica (Xueyang) Zhao

2029516888 | ericazhaoxy@gmail.com | LinkedIn: [ericazhaoxy](#) | GitHub: [ericazhaoxy](#) | ericazhaoxy.com

WORK EXPERIENCE

Machine Learning Engineer

MyEdMaster, LLC

Jul 2025 – Present

Remote

- Built and iterated multi-step LLM workflows for an adaptive tutoring assistant, improving response consistency across real student tasks. Designed an evaluation rubric and error taxonomy to diagnose hallucinations/omissions/irrelevance, enabling faster iteration on prompts and retrieval logic.
- Implemented backend services in Python (Flask/Django) to structure model outputs and integrate them into the product pipeline. Collaborated with cross-functional teammates to ship weekly improvements; tracked quality with internal test sets and user-feedback loops.

Graduate Instructional Assistant

George Washington University

Aug 2024 – May 2025

Washington, DC

- Supported data warehousing/ visualization courses operations for more than 60 students, including non-technical background ones: grading, office hours, and technical guidance on SQL and Tableau.
- Standardized grading rubrics and feedback templates to improve turnaround time and consistency.

PROJECTS

Mental Health Demand-Supply GapSight | Python, Pytrends, XGBoost, SHAP | [GitHub](#) Jan – May 2025

- Quantified mental health demand-supply gaps across U.S. states by building 2020–2024 state-month dataset integrating Google Trends (400+ mental-health queries) and HRSA/AHRF provider-capacity metrics.
- Trained interpretable machine learning models (XGBoost + SHAP) for anxiety/depression demand indices.
- Shipped interactive dashboard to track demand-supply gaps and top drivers by state and month. ([demo](#))

NIH Grant Activity Classifier | Python, scikit-learn, pandas, Classification | [GitHub](#) Aug – Dec 2024

- Built multi-class NIH grant activity classifier (F/T/K/R) using NIH RePORTER FY18–FY23 (118k+ records).
- Engineered labels from Activity Codes and harmonized cross-year features to support pipeline planning.
- Benchmarked 4 models with GridSearchCV; Histogram-based Gradient Boosting achieved 0.93 macro-F1 on validation set, outperforming Random Forest, Logistic Regression, and MLP baselines.

Personal Finance Tracker | React/Next.js, Node.js, PostgreSQL | [GitHub](#) Jan – Jun 2024

- Built a full-stack budgeting & expense tracking web app with PostgreSQL schema design and REST APIs.
- Delivered key features (CRUD, categorization, summaries) and deployed an end-to-end product.

TECHNICAL SKILLS

Languages | Python, JavaScript, Java, SQL, R

ML/GenAI | OpenAI, Gemini, Mistral AI, Hugging Face, TensorFlow

Frameworks | React, Django, Flask

Tools | Git, Docker, VSCode, PyCharm, PostgreSQL

Libraries | Pandas, NumPy, Matplotlib, Bootstrap

EDUCATION

M.S. Data Science

George Washington University

CCAS Dean's Award — \$13K Tuition Fellowship

May 2025

Washington, DC

M.S. Management

The University of Tokyo

March 2021

Tokyo, Japan

B.S. Management Information Systems (MIS)

Josai International University

Aug 2017

Tokyo, Japan